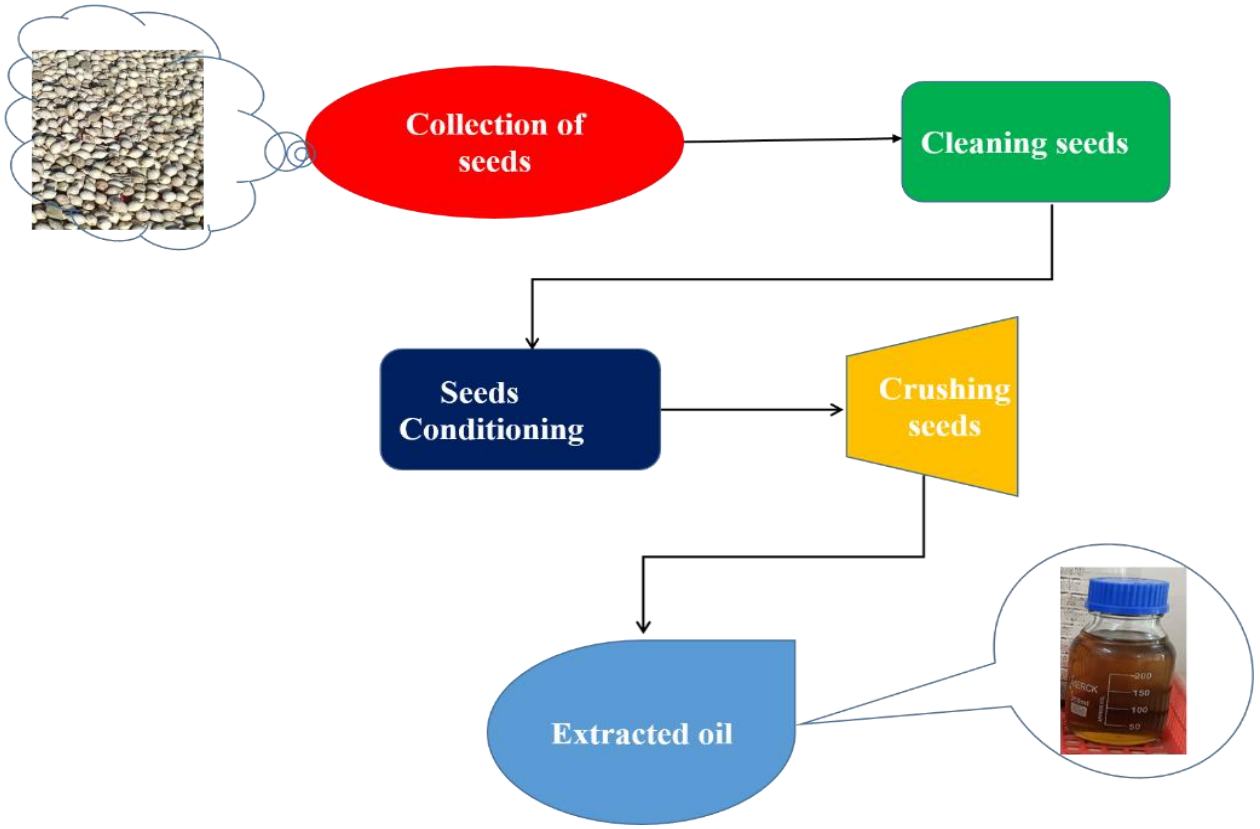
LINEAR PROGRAMMING

Thirty-first Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



What is an operation?

- An operation is some step in the overall process of producing a product or service that leads to the final output. (Example of extracting oil from seed.)



What is operations research?

- Operations Research (OR) started with World War II (1939–1945).
- The term was used by J.F. McClosky and F.N. Trefethen in 1940 in UK (?).
- In 1952, the Operations Research Society of America came into existence.
- Modern operational research originated at the Bawdsey Research Station in the UK in 1937 as the result of an initiative of the station's superintendent, A. P. Rowe and Robert Watson-Watt.
- 1949: OR unit was established in Regional Research Laboratory at Hyderabad. Another group was set up in the Defense Science Laboratory to solve the problems of stores, purchase and planning.
- 1953: OR unit was established at Indian Statistical Institute, Kolkata. Purpose: Use OR in national planning and survey.
- 1957: OR Society of India. It is a member of the International Federation of OR societies.

Definitions

- Operations research is the application of the methods of science to complex problems in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense. The distinctive approach is to develop a scientific model of the system incorporating measurements of factors such as chance and risk, with which to predict and compare the outcomes of alternative decisions, strategies or controls. The purpose is to help management in determining its policy and actions scientifically. (Operations Research Society, UK)
- The application of the scientific method to the study of large complex organizations or activities. It provides top level administrators with a quantitative basis for decision that will increase the effectiveness of such organizations in carrying out their basic purpose. (Committee on OR National Research Council, USA)

Typical Optimization Problem:

Minimize $f_1(x_1, x_2, \ldots, x_n)$, $f_2(x_1, x_2, \ldots, x_n)$, $\ldots\ldots\ldots$, $f_p(x_1, x_2, \ldots, x_n)$

Subject to

$$g_1(x_1, x_2, \ldots, x_n) \le 0$$

$$g_2(x_1, x_2, \ldots, x_n) \le 0$$

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$$g_m(x_1, x_2, \ldots, x_n) \le 0$$

$$h_1(x_1, x_2, \ldots, x_n) = 0$$

$$h_2(x_1, x_2, \ldots, x_n) = 0$$

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$$h_k(x_1, x_2, \ldots, x_n) = 0$$

$$x_i^l \le x_i \le x_i^u \quad \text{for } i = 1 \text{ to } n$$

Some points

- If there is only one objective function, the problem is called single objective optimization problem.
- If there are no constraints, the problem is called unconstrained optimization.
- The maximization of a function $f(\boldsymbol{x})$ is equivalent to minimization of $-f(\boldsymbol{x})$.

Operations research includes applied scientific methods and techniques to decision making problems for finding out the best or optimal solution.

Mathematical programming techniques are useful in finding out the minimum of a function of several variables under a prescribed set of constraints.

Operations research uses optimization techniques.

Optimization techniques find application not only in operations research but in many other areas as well, for example in finding out the zeros of a function.

Types of optimization problems

- (1) Linear Programming: Objective function and constraints are linear.
- (2) Dynamic programming problems: Determining the optimum solution to an *n*-variable problem by decomposing it into *n*-stages with each stage consisting of a single variable sub problem.
- (3) Integer programming: The variables are integers.
- (4) Nonlinear Programming: Constraints and/or objective functions are non-linear.
- (5) Goal programming: Converting multi-objective problems to single goals.

Pareto-optimal solution

Multi-objective optimization problems give rise to a set of optimal solutions known as Pareto-optimal solutions all of which are equally important as far as all objectives are concerned. In a Pareto-optimal set no one solution is better than the other in all respects, i.e. no one solution dominates the other.

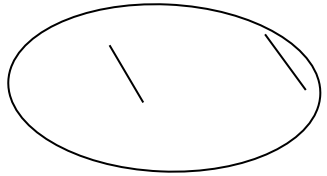
Some definitions

- **Cost function:** A function that is minimized is called the cost function.
- **Feasible solution:** All the constraints are satisfied.
- **Infeasible solution:** One or more constraints are not satisfied.
- **Combinatorial optimization:** Finding an optimal object from a finite set of objects.

  Example: Travelling salesman problem.
- **Discrete optimization:** (1) Integer programming
- (2) Combinatorial optimization

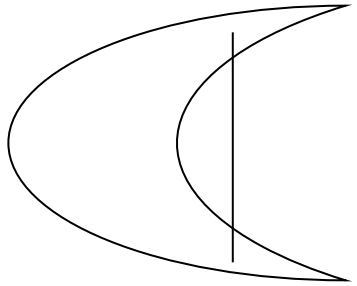
Convex Set

- A set C is convex if every point on the line segment connecting *x* and *y* is in C.



The convex hull of a set of points X is the minimal convex set connecting X.

Non-convex Set



Linear Programming Example

- Problem: Product M1 requires 1 kg R_1 and 2 kg R_2. One unit of M_2 requires 3 kg R_1 and 1 kg R_2. Maximum available R_1 is 100 kg and maximum available R_2 is 150 kg. Profit of $M_1$is Rs2/unit and that of M_2 is Rs. 3/unit. Formulate the problem.
- Solution:

Maximize $2 M_1 + 3 M_2$

Subject to

$M_1 + 3 M_2 \leq 100$

$2M_1 + M_2 \leq 150,$

$M_1, M_2 \geq 0$

Problem: Factory Production Planning

A factory produces **three products**: P1, P2, and P3. Let the production quantities of these products be the **design variables**:

- x_1: number of units of P1 to produce
- x_2: number of units of P2 to produce
- x_3: number of units of P3 to produce

The objective is to **maximize profit**.

Objective Function:

The profit per unit is as follows:

- P1: $30
- P2: $20
- P3: $40

So, the objective function is:

$$\text{Maximize } Z = 30x_1 + 20x_2 + 40x_3$$

Constraints:

1. **Machine Time (≤ type):**

   Each product requires some machine time per unit:

- P1: 2 hours
- P2: 1 hour
- P3: 3 hours

Total machine time available: **100 hours**

$$2x_1 + x_2 + 3x_3 \leq 100$$

2. **Labor Availability (≥ type):**

   The factory must use **at least 40 hours** of labor. Labor hours per product:

- P1: 1 hour
- P2: 2 hours
- P3: 1 hour

$$x_1 + 2x_2 + x_3 \geq 40$$

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3. **Material Balance (= type):**

   To maintain balance in inventory, **the number of units of P1 must equal P3**:

$$x_1 = x_3$$

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4. **Non-negativity (bound constraints):**

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0$$

5. **Maximum market demand (upper bounds):**

- $x_1 \leq 40$
- $x_2 \leq 50$
- $x_3 \leq 30$

Summary (LP Formulation):

$$\text{Maximize } Z = 30x_1 + 20x_2 + 40x_3$$

Subject to:

$$\begin{gathered} 2x_1 + x_2 + 3x_3 \leq 100 \quad \text{(Machine time constraint)} \\ x_1 + 2x_2 + x_3 \geq 40 \quad \text{(Labor constraint)} \\ x_1 - x_3 = 0 \quad \text{(Inventory balance)} \\ 0 \leq x_1 \leq 40 \\ 0 \leq x_2 \leq 50 \\ 0 \leq x_3 \leq 30 \end{gathered}$$

Standard Linear Program

Minimize $c_1x_1 + c_2x_2 + \ldots\ldots\ldots\ldots\ldots + c_nx_n$

Subject to

$a_{11}x_1 + a_{12}x_2 + \ldots\ldots\ldots\ldots\ldots + a_{1n}x_n = b_1$

$a_{21}x_1 + a_{22}x_2 + \ldots\ldots\ldots\ldots\ldots + a_{2n}x_n = b_2$

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$a_{m1}x_1 + a_{m2}x_2 + \ldots\ldots\ldots\ldots\ldots + a_{mn}x_n = b_m$

$x_1 \geq 0,\ x_2 \geq 0, \ldots\ldots\ldots\ldots\ldots, x_n \geq 0$

We assume that each $b_i \geq 0$.

In compact form:

Minimize $\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x}$

subject to $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}$

Slack and surplus variables may be used for converting inequalities into equalities.

Free variables can be expressed as the difference of 2 non-negative variables.

Basic solutions: Consider a linear programming problem consisting of n decision variables and m equality constraints. A solution obtained after making $(n-m)$ variables 0 is called basic solution.

Basic variables: If there are m equality constraints in a linear programming problem having n decision variables, then we can assign 0 values to $(n-m)$ variables and solve the m equations for remaining m variables. These remaining m variables are called basic variables.